



Sample Java Code and Junit Test Script

Java Code:

```
class EmployeeDetails {  
    private String name;  
    private double monthlySalary;  
    private int age;  
    public String getName() {  
        return name;  
    }  
    public void setName(String name) {  
        this.name = name;  
    }  
    public double getMonthlySalary() {  
        return monthlySalary;  
    }  
    public void setMonthlySalary(double monthlySalary) {  
        this.monthlySalary = monthlySalary;  
    }  
  
    public int getAge() {  
        return age;  
    }  
}
```

```

    }

    public void setAge(int age) {
        this.age = age;
    }
}

public class EmpBusinessLogic {

    public double calculateYearlySalary(EmployeeDetails
employeeDetails){
        double yearlySalary=0;
        yearlySalary = employeeDetails.getMonthlySalary() * 12;
        return yearlySalary;
    }

    public double calculateAppraisal(EmployeeDetails employeeDetails){
        double appraisal=0;
        if(employeeDetails.getMonthlySalary() < 10000){
            appraisal = 500;
        }else{
            appraisal = 1000;
        }
        return appraisal;
    }
}

```

Compile the above program and create a class file

Example set path= choose the jdk bin directory

```
javac EmpBusinessLogic.java
```

Download junit3.8.1 executable JAR file and extract it

Set classpath to junit3.8.1.jar folder

(example SET CLASSPATH=.;%CLASSPATH%; Location of the jar file

example: E:\junit\junit3.8.1.jar

Type the following Junit Test Scripts to test the above Java code

```
import junit.framework.TestCase;
```

```
public class TestEmployeeDetails extends TestCase
```

```
{
```

```
    EmpBusinessLogic empBusinessLogic =new EmpBusinessLogic();
```

```
    EmployeeDetails employee = new EmployeeDetails();
```

```
//test to check appraisal
```

```
public void testCalculateAppraisal()
```

```
{
```

```
    employee.setName("Rajeev");
```

```
    employee.setAge(25);
```

```
    employee.setMonthlySalary(8000);
```

```

        double appraisal= empBusinessLogic.calculateAppraisal(employee);
        assertEquals(500, appraisal, 0.0);
    }

    // test to check yearly salary

    public void testCalculateYearlySalary()
    {
        employee.setName("Rajeev");
        employee.setAge(25);
        employee.setMonthlySalary(8000);
        double salary= empBusinessLogic.calculateYearlySalary(employee);
        assertEquals(96000, salary, 0.0);
    }
}

```

Save the above Junit Test Script as TestEmployeeDetails.java and compile the program using JAVAC compiler

```
javac TestEmployeeDetails.java
```

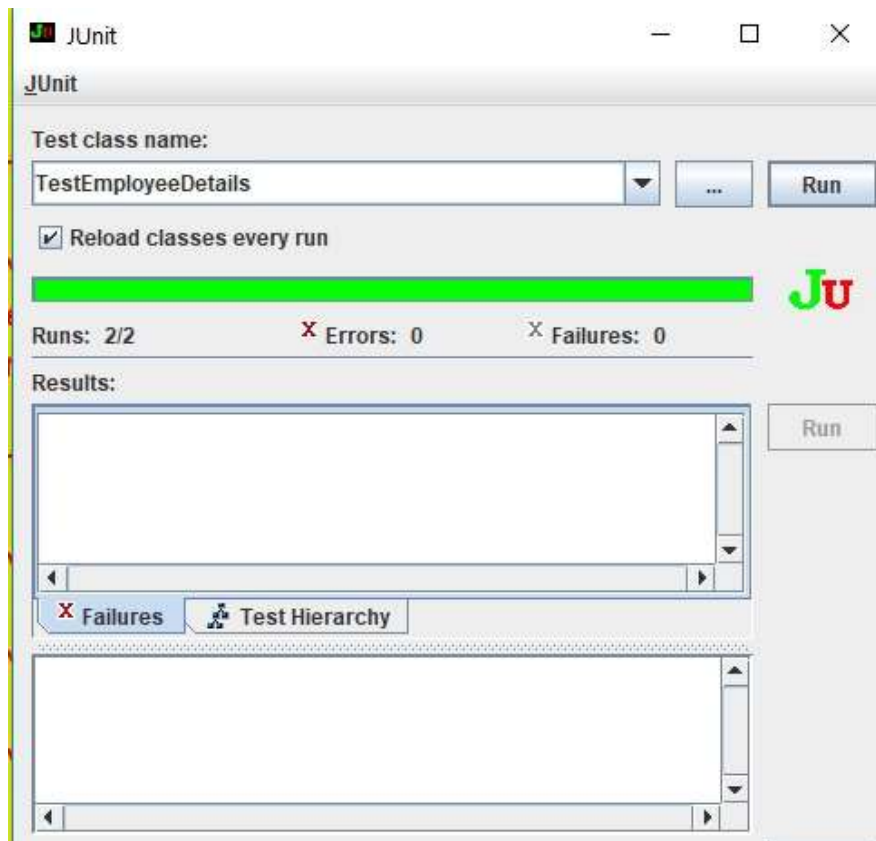
After Successful compilation TestEmployeeDetails.class will be created

To see the test result type the following command

```
java junit.swingui.TestRunner TestEmployeeDetails
```

You will get the following User Interface to see the test result.

1. If you get green progress bar, indicates the test result is successful.
2. If you get red progress bar, indicates the test result is unsuccessful (Either it could be coding error or it could be test script logical error).



2nd Programme:

```
public class Calculator {  
    public static int add(int number1, int number2) {  
        return number1 + number2;  
    }  
  
    public static int sub(int number1, int number2) {  
        return number1 - number2;  
    }  
  
    public static int mul(int number1, int number2) {  
        return number1 * number2;  
    }  
  
    // Integer divide. Return a truncated int.  
    public static int divInt(int number1, int number2) {  
        if (number2 == 0) {  
            throw new IllegalArgumentException("Cannot divide by 0!");  
        }  
        return number1 / number2;  
    }  
}
```

```
// Real number divide. Return a double.

public static double divReal(int number1, int number2) {
    if (number2 == 0) {
        throw new IllegalArgumentException("Cannot divide by 0!");
    }
    return (double) number1 / number2;
}
}
```

JUnit Test Script Code:

(only add method is tested, write additional test scripts to test other methods of calculator class)

```
import junit.framework.TestCase;
```

```
public class TestCalculator extends TestCase
{
    Calculator c = new Calculator();
```

```
//test to check add
```

```
    public void testAdd()
    {
        int d=c.add(10,20);
```

```
        assertEquals(30,d, 0.0);
```

```
d = c.add(20,30);  
assertEquals(50,d,0.0);  
d=c.add(-5,4);  
assertEquals(-2,d,0.0);  
  
}
```

Compile and run the Java code as well as Junit test script and analyse the test result.